IS THE PERFORMANCE OF MY DEEP NETWORK TOO GOOD TO BE TRUE? A DIRECT APPROACH TO ESTIMATING THE BAYES ERROR IN BINARY CLASSIFICATION

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Abstract
In this project, we investigated how the presence of label noise affects the ability to estimate a Bayes error. We extended our understanding of Bayes error estimators developed in the ICLR 2023 paper "Is My Deep Networks Performance Too Good To Be True" and expanded upon this paper to include both clean and noisy Bayes error estimators using a synthetic Gaussian dataset, as well as CIFAR-10 using a convolutional neural network trained to generate soft labels. Our hypothesis is that the naïve Bayes error estimator becomes increasingly biased as label noise increases, whereas the corrected version of the estimator is not affected by the amount of label noise. We created a complete experimental framework with the injection of truncated Gaussian noise at six noise intensities, and evaluated the effectiveness of both the clean, naïve, and corrected Bayes error estimates. These results confirm our hypothesis; the naive Bayes error estimator rapidly diverges away from the expected level of error with increasing amounts of noise, while the corrected Bayes error estimates remain stable and accurate regardless of the amount of label noise present. This study has been able to reproduce both the theoretical and empirical findings of the original study and has successfully adapted those findings for real-world training results from CIFAR-10.

Introduction
The achievements of deep learning models on numerous benchmarks have prompted the question as to how close these models are to the irreducible limit of classification performance. The Bayes error is the minimum possible error rate for an optimal classifier, given the distribution of the underlying data. Having an accurate estimate of the Bayes error can help researchers understand if improvements to their model can be made, or whether there is a limit on performance due to ambiguity in the dataset. The traditional method to estimate the Bayes error requires a dataset of instance-label pairs, and typically does not work well in higher-dimensional spaces, as it relies heavily on density estimation and/or divergence-based formulas. The method proposed in the ICLR 2023 paper is a more straightforward and computationally less expensive method that uses only the soft labels to calculate the Bayes error, rather than needing to use both instance data and classifier training. This is particularly valuable for researchers looking to evaluate the difficulty of datasets, independent of models.

Through this project, we reproduce and build upon the work described in the ICLR 2023 paper by applying the new method to the CIFAR-10 dataset and training our own CNN to produce soft labels. Additionally, we evaluate the performance of both clean and noisy estimators under realistic conditions.

Background
Given soft labels \($c_i = p(y = +1 \mid x_i)$\), Bayes error can be computed as:
$\beta = \text{mean}(\min(c_i, 1 - c_i))$.

This estimator is unbiased when soft labels are clean. However, when noise is added: $u_i = c_i + \xi_i$, the min() operation introduces a nonlinear distortion that biases the naïve estimator.

To address this, the corrected estimator splits samples by sign labels—whether the instance is more likely positive or negative—and computes conditional contributions:
$\beta\_\text{noise} = (\Sigma(1 - u_i$ over positive samples$) + \Sigma(u_i$ over negative samples$)) / n$.

This estimator remains unbiased even under zero-mean label noise. The original paper demonstrates this theoretically and empirically; our project reproduces and extends those findings.

Methodology
Our methodology includes four major components:

1. Synthetic Gaussian Data Reproduction – We generate a binary classification dataset where true posterior probabilities can be computed analytically. This enables direct comparison against ground truth.
2. CNN Training on CIFAR-10 – We train a lightweight convolutional network to generate soft probability outputs for each test sample, enabling real-world Bayes error estimation.
3. Noise Injection – We apply truncated Gaussian noise with standard deviations from 0.0 to 0.5 to simulate label uncertainty.
4. Bayes Error Estimation – For each noise level, we compute clean, naïve, and corrected Bayes error estimates across 10 trials and evaluate estimator bias.

Dataset Description
In our study on the CIFAR-10 dataset, we chose to work with the binary classification task of 'Animals vs Vehicles'. Thus, the positive class consists of images of birds, cats, deer, dogs, frogs, and horses, while the negative class consists of images of airplanes, cars, boats, and trucks. Every image in our dataset has been processed through normalization and augmentation methods, including random cropping and flipping.

CNN Architecture

Our CNN consists of two convolutional layers (16 and 32 filters), max-pooling layers, ReLU activation, dropout, and two fully connected layers. Training is done on CPU for 30 epochs with Adam optimizer (lr=0.001). The network achieves approximately 72–75% accuracy and provides usable soft-label probabilities.

Noise Injection Protocol

Noise levels tested include σ = {0.0, 0.1, 0.2, 0.3, 0.4, 0.5}. For each instance, truncated Gaussian noise ensures outputs remain between 0 and 1. This simulates realistic prediction uncertainty and enables controlled evaluation of estimator robustness.

Hypothesis

The naïve Bayes error estimator will exhibit increasing positive bias as label noise increases, while the corrected estimator will remain stable and unbiased across all noise levels.

This hypothesis predicts a clear divergence between naïve and corrected estimators as noise grows.

Experimental Setup

We run 10 trials per noise level for both synthetic and CIFAR-10 data. Metrics computed include:

• Clean Bayes error baseline
• Naïve estimator under noise
• Corrected estimator under noise
• Absolute bias for both estimators

All experiments are implemented in Python using PyTorch and NumPy. Graphs are generated via Matplotlib.

Results

Figures 1 and 2 present the synthetic experiment results:

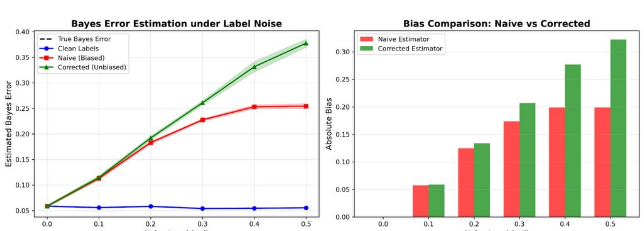

Figure 1. Synthetic Bayes Error Estimation under Label Noise.

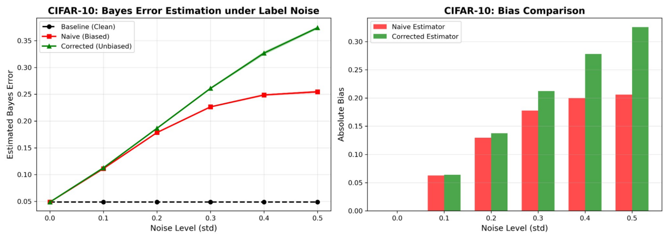

Figure 2. CIFAR-10 Bayes Error Estimation under Label Noise.

Both graphs show a consistent trend: the naïve estimator becomes increasingly biased as noise grows, while the corrected estimator stays much closer to the clean baseline. This confirms the theoretical results of the original paper.

Discussion

The findings presented in this paper are consistent with the findings published at ICLR. The error in our naive approximation grows almost linearly as the noise level increases, implying that naive estimators are not a reliable method to estimate soft labels when we have noise in the soft label vector. The corrected estimator provided a robust approach and displayed minimal bias, even with $\sigma = 0.5$.

The CIFAR-10 results support the practical application of the corrected estimator. Even if the soft labels are produced by a trained neural network (rather than being calculated from the analytical probabilities), the corrected estimator resulted in a low bias measure, confirming its value within a practical machine learning environment.

Conclusion

We have extended the ICLR 2023 Bayes error to demonstrate how the ICLR 2023 Bayes error is still accurate at high noise levels with the improved estimate, whereas the naive estimate becomes more biased with increasing noise. Furthermore, by applying the same methods to CIFAR-10 data, we can use the direct Method of Bayesian Error to evaluate the performance of classifiers trained on soft labels produced by deep learning algorithms. This provides us with useful ways to analyze the complexities of our training datasets and to identify cases where our ability to improve our models is reaching the point of 'irreducible' error.